Technical Summary:
Boosting Language Models Reasoning with Chain-of-Knowledge Prompting

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1 Research Problem and Motivation

1.1 Research Problem

The paper studies how to improve the reasoning ability of large language models (LLMs) on complex tasks such as commonsense question answering, factual verification, symbolic manipulation, and arithmetic word problems. The central problem is that standard Chain-of-Thought (CoT) prompting often produces reasoning chains that are fluent and plausible in natural language but contain factually incorrect statements or unfaithful explanations, which leads to unreliable final answers.

1.2 Motivation and Research Gap

Chain-of-Thought prompting encourages LLMs to generate step-by-step rationales before producing a final answer. This strategy has substantially improved performance on many multi-step reasoning benchmarks. However, two core issues remain:

- **Factuality problem.** CoT-style rationales frequently contain hallucinated or incorrect intermediate facts. For example, in a sports plausibility task, a rationale may incorrectly claim that a specific player is associated with a sport that the player has never played, and this hallucinated fact is then used to justify the final answer.
- **Faithfulness problem.** The generated explanation may not faithfully reflect the actual internal computation that produced the answer. A model can output a coherent explanation while still predicting an incorrect answer, which suggests that the explanation is more of a post-hoc justification than a faithful trace of reasoning.

These issues expose a gap between current CoT prompting and the goal of trustworthy reasoning. Prior work on in-context learning and CoT mainly focuses on *eliciting* rationales but provides limited machinery to *verify* their factual correctness and faithfulness.

The paper identifies several research gaps:

- Reasoning evidence in CoT remains unstructured, which makes it difficult to align with external knowledge bases for fine-grained verification.
- Existing methods seldom combine factuality and faithfulness into a unified and quantitative reliability score for individual reasoning chains.
- Few methods use verification feedback to *actively* revise and improve reasoning, especially by injecting explicit, correct knowledge and prompting the model to rethink.

2 Related Work

2.1 In-Context Learning and Standard Prompting

In-Context Learning (ICL) enables LLMs to perform new tasks by conditioning on a few demonstration examples in the prompt, without parameter updates. Standard prompting uses input-output pairs as exemplars and asks the model to directly predict answers for new queries. This strategy is effective for many classification and generation tasks but often struggles on multi-step reasoning problems that require explicit intermediate steps.

2.2 Chain-of-Thought Prompting

Chain-of-Thought prompting extends standard prompting by including intermediate reasoning steps in the demonstrations. Typical variants include:

- **Manual CoT**, where human-written rationales are added to exemplars.
- **Zero-Shot CoT**, which uses a prompting phrase such as "Let us think step by step" to elicit rationales for new queries without labeled explanations.
- **Auto-CoT**, which automatically selects exemplars and their rationales based on heuristics.
- **ComplexCoT**, which uses complex exemplars to encourage deeper reasoning.
- **Self-Consistency (SC)**, which samples multiple reasoning paths and chooses the final answer by majority vote across sampled rationales.

These methods substantially improve accuracy on reasoning benchmarks but largely treat the generated rationales as opaque text, without explicit structures for verification against external knowledge.

2.3 Retrieval-Augmented Reasoning and Verification

Retrieval-augmented methods enrich model inputs with external evidence, for example by querying knowledge bases or text corpora. Some work designs verifiers for specific domains such as math word problems or uses supervised signals to train auxiliary models that check the correctness of intermediate steps.

However, these approaches often lack:

- A structured representation of reasoning evidence that can be aligned with knowledge graphs.
- A joint treatment of factual correctness and faithfulness in a single reliability metric.
- A generic rethinking mechanism that uses verification feedback to iteratively improve reasoning across diverse tasks.

3 Dataset Construction

3.1 Reasoning Benchmarks

The paper evaluates the proposed method on three categories of benchmarks. All tasks use accuracy as the evaluation metric, defined as the fraction of test questions for which the predicted answer exactly matches the ground-truth answer.

- **Commonsense and factual reasoning:**
  - **CommonsenseQA (CSQA).** A multiple-choice dataset that tests commonsense reasoning beyond the literal text.
  - **StrategyQA.** A yes/no question answering dataset where many questions require multi-hop commonsense reasoning.
  - **OpenBookQA.** A multiple-choice science QA dataset where questions must be answered using a small "open book" of science facts combined with commonsense.
  - **ARC-challenge (ARC-c).** The challenging subset of the AI2 Reasoning Challenge, consisting of difficult science exam questions.
  - **Sports Understanding.** Questions about sports scenarios and rules that require domain-specific commonsense.
  - **BoolQ.** A yes/no reading comprehension dataset with short passages and questions.
- **Symbolic reasoning:**
  - **Last Letter Concatenation (Letter).** Given a list of words, the task is to output the concatenation of their last letters.
  - **Coin Flip (Coin).** Given a description of a sequence of coin flips and operations, the task is to predict the final state of the coin.
- **Arithmetic reasoning:**
  - **GSM8K.** Grade-school math word problems requiring multi-step arithmetic reasoning.
  - **SVAMP.** Math word problems that are constructed to test robustness and reasoning beyond surface patterns.
  - **AQuA.** Algebraic word problems presented in multiple-choice format.
  - **MultiArith.** Arithmetic word problems involving multi-step integer operations.

A concise overview is presented in Table 1.

3.2 Knowledge Bases for Factuality Verification

The method relies on external knowledge bases (KBs) to verify the factuality of model-generated evidence. The paper aggregates knowledge from multiple sources, each covering different aspects:

- **Dictionary-like knowledge.** Wiktionary and similar resources provide word-level information such as meanings, parts of speech, and morphological attributes.
- **Commonsense knowledge.** ConceptNet encodes relations such as *IsA*, *PartOf*, and *UsedFor* between common concepts.
- **Entity knowledge.** Large-scale entity-centric knowledge graphs such as Wikidata5M store facts about real-world entities.
- **Event and script knowledge.** Knowledge bases such as ATOMIC, GLUCOSE, ASER, and CausalBank capture event-level and causal relations, scripts, and typical event sequences.

All knowledge is represented as triples of the form $(\text{subject}, \text{relation}, \text{object})$. For the Last Letter task, a dedicated dictionary-like KB is constructed in which each triple links a word to its last letter. For arithmetic reasoning and the Coin Flip task, suitable structured KBs are not available; therefore, the factuality verification component is not applied to these tasks.

Group	Dataset	Task Type	Answer Format
Commonsense & factual	CSQA	Commonsense multiple-choice QA	Option label
	StrategyQA	Implicit multi-step commonsense QA	Yes / No
	OpenBookQA	Science QA with open-book facts	Option label
	ARC-c	Challenging science exam QA	Option label
	Sports	Sports rules and scenarios QA	Option label
	BoolQ	Passage-based QA	Yes / No
Symbolic	Letter	Last-letter concatenation	Character sequence
	Coin	Coin flip state tracking	Final state token
Arithmetic	GSM8K	Grade-school math word problems	Numeric answer
	SVAMP	Robust math word problems	Numeric answer
	AQuA	Algebraic word problems	Option label
	MultiArith	Multi-step arithmetic problems	Numeric answer

Table 1: Datasets used for evaluating the proposed Chain-of-Knowledge prompting and verification framework. All tasks are evaluated using accuracy.

3.3 Construction of Chain-of-Knowledge Exemplars

To construct in-context demonstrations for Chain-of-Knowledge (CoK) prompting, the paper follows a three-step process for each task:

1. Randomly select K labeled questions as basic exemplars, following common practice in few-shot CoT prompting.
2. For each selected question, automatically generate a textual rationale using zero-shot CoT, with a prompt such as "Let us think step by step". This rationale serves as an *explanation hint*.
3. Retrieve candidate knowledge triples from the aggregated KB based on the concepts appearing in the question and the explanation. Then, human annotators refine and select a final list of evidence triples that support the reasoning for the exemplar.

Each exemplar is thus represented as a quadruple (Q_i, T_i, H_i, A_i), where Q_i is the question, T_i is a list of evidence triples, H_i is the explanation hint, and A_i is the final answer.

4 Query Protocol and Task Definitions

4.1 Notation

Let $E = \{(Q_i, T_i, H_i, A_i)\}_{i=1}^{K}$ denote the set of Chain-of-Knowledge exemplars, where:

- Q_i is the i-th exemplar question.
- $T_i = \{(s_{ij}, r_{ij}, o_{ij})\}_j$ is the list of evidence triples, with subject s_{ij}, relation r_{ij}, and object o_{ij}.

- H_i is the natural language explanation hint.
- A_i is the correct answer for Q_i.

Given a test query $\hat{Q}_i$, a prompt is formed by concatenating all exemplars in E and the test query. Due to the CoK format in the exemplars, the model is encouraged to output three components for each test query:

- a list of evidence triples $\hat{T}_i$,
- a natural language explanation hint $\hat{H}_i$,
- and a final answer $\hat{A}_i$.

4.2 Task Definitions

The paper considers the following task settings:

- For commonsense and factual reasoning tasks, the model must output a correct option label or a yes/no answer, supported by evidence triples and explanations.
- For symbolic reasoning tasks, the model must output an exact symbolic sequence, such as a concatenation of characters for the Last Letter task, or a token indicating the final coin state.
- For arithmetic reasoning tasks, the model must output a numeric answer that matches the ground-truth integer.

In all cases, the explanation and triples are used internally by the proposed verification mechanism and are not directly supervised by task labels.

5 Modeling Approach

The modeling framework contains three major components:

1. Chain-of-Knowledge prompting, which explicitly elicits structured knowledge evidence and explanations.
2. F2-Verification, which quantifies the reliability of a reasoning chain in terms of factuality and faithfulness.
3. A rethinking process, which iteratively refines unreliable reasoning by injecting correct knowledge.

5.1 Chain-of-Knowledge Prompting

5.1.1 Evidence Triples (CoK-ET)

The core idea is to treat reasoning as a process over an explicit knowledge system. Each piece of evidence is represented as a triple (s_{ij}, r_{ij}, o_{ij}):

- s_{ij} is the subject entity, for example the name of a person or an object.
- r_{ij} is the relation, such as *isA*, *part of*, or *is commonly used in*.
- o_{ij} is the object entity, such as a category, an event, or another object.

A list of such triples T_i forms an explicit knowledge map that links the question to the answer.

5.1.2 Explanation Hints (CoK-EH)

The explanation hint H_i is a textual rationale similar to a CoT explanation. It describes how the evidence triples in T_i are combined to reach the final answer A_i. For example, it may state that a particular action is typical in one sport but not in another, and therefore a given sentence is implausible.

5.1.3 Prompt Format

The CoK prompt is designed so that each exemplar presents the pattern:

$$\text{Question} \rightarrow \text{Evidence Triples} \rightarrow \text{Explanation Hint} \rightarrow \text{Answer.}$$

For a new query, the model is expected to follow the same pattern, first generating explicit evidence triples, then a grounded explanation, and finally an answer. Compared with plain CoT, this format makes reasoning more structured and more amenable to subsequent verification.

5.2 F2-Verification: Factuality and Faithfulness

5.2.1 Factuality Verification

Factuality measures whether the generated evidence triples agree with the external knowledge base K. For each generated triple $(\hat{s}_{ij}, \hat{r}_{ij}, \hat{o}_{ij})$ in $\hat{T}_i$, the paper defines a factuality scoring function

$$fv(\hat{r}_{ij} \mid \hat{s}_{ij}, \hat{o}_{ij}, K),$$

which outputs a real-valued score reflecting how plausible the triple is under K.

Two strategies are used:

- **Exact verification.** When a triple exactly appears in the KB, factuality can be obtained from a binary indicator that checks whether $(\hat{s}_{ij}, \hat{r}_{ij}, \hat{o}_{ij})$ is in K. This captures cases where the fact is explicitly stored.
- **Implicit verification.** Many plausible facts are not explicitly present in K. To estimate plausibility in such cases, the paper uses the TransR knowledge graph embedding model. TransR embeds entities and relations as vectors and assigns lower energy to plausible triples and higher energy to implausible ones. The factuality score is derived from this energy, after normalization, so that more plausible triples receive higher scores.

The exact and implicit strategies are combined, so that explicit matches are used when available, and TransR-based scores are used otherwise.

5.2.2 Faithfulness Verification

Faithfulness measures whether the explanation is consistent with the question, evidence, and answer. For a test query $\hat{Q}_i$, a list of evidence triples $\hat{T}_i$, and a final answer $\hat{A}_i$, the paper constructs a concatenated sequence

$$\hat{H}'_i = [\hat{Q}_i; \hat{T}_i; \hat{A}_i].$$

The original explanation $\hat{H}_i$ and the constructed sequence $\hat{H}'_i$ are encoded using SimCSE, a sentence encoder trained by contrastive learning. SimCSE maps sentences into a vector space in which semantically similar sentences are close. The faithfulness score is defined as

$$f_u(\hat{H}_i \mid \hat{H}'_i),$$

which is the similarity between the embeddings of $\hat{H}_i$ and $\hat{H}_i'$. A higher similarity indicates that the explanation is more aligned with the evidence and answer, and therefore more faithful.

5.2.3 Overall Reliability Score

The paper combines factuality and faithfulness into a single reliability score C_i for each test query. The reliability score is defined as

$$C_i = \gamma \cdot \frac{1}{|\hat{T}_i|} \sum_{j=1}^{|\hat{T}_i|} fv(\hat{r}_{ij} \mid \hat{s}_{ij}, \hat{o}_{ij}, K) + (1-\gamma) \cdot f_u(\hat{H}_i \mid \hat{H}_i'), \tag{1}$$

where:

- C_i lies between 0 and 1 and reflects how reliable the reasoning chain is.
- $|\hat{T}_i|$ is the number of generated evidence triples for the i-th query.
- $fv(\hat{r}_{ij} \mid \hat{s}_{ij}, \hat{o}_{ij}, K)$ is the factuality score of the j-th triple, after normalization so that more plausible triples receive larger values.
- $f_u(\hat{H}_i \mid \hat{H}_i')$ is the faithfulness score defined via SimCSE similarity.
- $\gamma \in (0, 1)$ is a balancing factor between factuality and faithfulness, set to 0.5 in the experiments.

Intuitively, the first term averages how factually correct the evidence triples are, and the second term measures how well the explanation aligns with the question, evidence, and answer.

5.3 Rethinking Process

The rethinking process uses the reliability score C_i to iteratively refine predictions.

A reliability threshold $\theta \in (0, 1)$ and a maximum iteration number $N \geq 1$ are defined. All test queries initially belong to an unreliability set U, indicating that their reasoning chains have not yet been verified.

For each iteration $n = 1, \ldots, N$:

1. For each query $\hat{Q}_i \in U$, construct a CoK prompt. In the first iteration, this prompt contains the original exemplars and the query. In later iterations, it may also include injected knowledge.
2. Run the LLM with the CoK prompt to generate evidence triples $\hat{T}_i^{(n)}$, explanation $\hat{H}_i^{(n)}$, and answer $\hat{A}_i^{(n)}$.
3. Compute the reliability score $C_i^{(n)}$ using Equation (1).
4. If $C_i^{(n)} \geq \theta$, treat the reasoning chain as reliable, fix $\hat{A}_i = \hat{A}_i^{(n)}$, and remove $\hat{Q}_i$ from U.
5. If $C_i^{(n)} < \theta$, keep the query in U. Identify the evidence triples with low factuality scores and retrieve correct triples from the KB to replace or augment them. Inject these corrective triples into the prompt for the next iteration.

After at most N iterations, any remaining query in U selects as final answer the $\hat{A}_i^{(n)}$ corresponding to the highest reliability score across iterations. This process resembles a student revising solutions after receiving feedback on which supporting facts are wrong.

6 Empirical Results

6.1 Experimental Setup

Models. The experiments primarily use two LLMs:

- **text-davinci-002**, a GPT-3 series model with approximately 175B parameters.
- **gpt-3.5-turbo**, a widely used conversational model.

Generation uses greedy decoding with temperature set to zero and maximum output length set to a fixed limit, aligned with baselines in prior work.

Baselines. The baselines include:

- Standard prompting (SP) in zero-shot and few-shot settings.
- Manual CoT with human-written rationales.
- Zero-shot CoT with a prompt such as "Let us think step by step".
- Auto-CoT and ComplexCoT as enhanced CoT-based prompting strategies.
- Self-Consistency (SC) applied to CoT variants, where multiple reasoning paths are sampled and the most frequent answer is selected.

In addition, the paper compares with supervised fine-tuning baselines, in which task-specific models are trained and evaluated on the same benchmarks.

6.2 Overall Performance

Across commonsense, factual, symbolic, and arithmetic reasoning tasks, Chain-of-Knowledge prompting yields consistent improvements over CoT-based baselines.

For **text-davinci-002**:

- CoK improves accuracy over strong CoT baselines on most datasets, typically by a few percentage points. The gains are particularly notable on commonsense and factual benchmarks such as CSQA, StrategyQA, ARC-c, and BoolQ.
- When the F2-Verification mechanism is enabled (CoK + F2-V), performance further improves on tasks that have suitable knowledge bases, and on some datasets the accuracy approaches that of fine-tuned models.

For **gpt-3.5-turbo**:

- CoK outperforms Manual CoT and ComplexCoT on many benchmarks, showing that the structured evidence layer is beneficial across different model families.
- Combining CoK with Self-Consistency (CoK + SC) produces the best or near-best results on several tasks, indicating that CoK is complementary to sampling-based aggregation of reasoning paths.

Overall, the quantitative results demonstrate that explicitly modeling evidence triples and explanations in the prompt leads to more accurate reasoning than purely textual CoT.

6.3 Ablation Studies

The paper performs ablation studies to analyze the contribution of each component.

Effect of evidence triples and explanation hints. Several CoK variants are evaluated:

- CoK without evidence triples (CoK w/o ET), where only explanation hints are used.
- CoK without explanation hints (CoK w/o EH), where only evidence triples are used.
- CoK + F2-V without factuality, where the reliability score ignores factuality.
- CoK + F2-V without faithfulness, where the reliability score ignores faithfulness.

The results show that:

- Removing either evidence triples or explanation hints leads to a noticeable drop in accuracy, confirming that both structured evidence and textual explanations are important.
- The full F2-Verification that combines factuality and faithfulness yields higher accuracy than variants that rely on only one of the two signals.

Robustness to noisy exemplars. The paper studies the effect of noise in CoK exemplars by corrupting a portion of evidence triples with incorrect ones. Accuracy decreases as the noise ratio increases, but the degradation is gradual, and the method remains relatively robust at moderate noise levels. This indicates that the model and verification mechanism can partly compensate for imperfect demonstrations.

6.4 Effect of F2-Verification and Rethinking

On tasks where knowledge bases are available, adding F2-Verification and the rethinking process provides additional gains over pure CoK prompting:

- The reliability score C_i helps identify instances where the initial reasoning is unreliable, and these instances benefit from knowledge injection and regeneration.
- The iterative refinement leads to improved accuracy on multiple commonsense and factual reasoning benchmarks.

On arithmetic datasets and the Coin Flip task, F2-Verification is disabled due to the lack of suitable KBs. In these cases, the improvements observed with CoK originate from the structured prompting itself rather than from verification.

7 Summary

7.1 Innovation and Comparison with Prior Methods

The main innovations of the paper can be summarized as follows:

- **Chain-of-Knowledge prompting.** Instead of relying solely on unstructured chain-of-thought explanations, the paper proposes a prompt format that explicitly elicits both evidence triples and explanation hints. This structured representation exposes the knowledge used in reasoning and enables fine-grained verification.

- **F2-Verification.** A new verification framework is introduced to quantify the reliability of reasoning chains by combining factuality (based on knowledge graph matching and TransR-based plausibility) and faithfulness (based on SimCSE similarity between explanations and the concatenation of question, evidence, and answer).
- **Rethinking with knowledge injection.** The framework iteratively refines unreliable reasoning by injecting correct knowledge triples into the prompt and prompting the LLM to rethink. This generate-evaluate-correct cycle reduces the impact of hallucinated or unfaithful reasoning.
- **Compatibility with existing prompting strategies.** The CoK format can be used together with Manual CoT, Zero-Shot CoT, Auto-CoT, ComplexCoT, and Self-Consistency. Experiments show that combinations such as CoK + SC often provide the strongest performance.

Compared with prior CoT-based methods that treat rationales as unstructured text, the proposed approach introduces an explicit knowledge layer and a principled verification mechanism, leading to more trustworthy reasoning.

7.2 Limitations and Future Directions

The paper also acknowledges several limitations:

- **Dependence on knowledge bases.** F2-Verification requires high-quality knowledge bases. For domains without suitable KBs, only the structural benefits of CoK are available, and factuality cannot be directly checked.
- **Manual annotation cost.** Constructing CoK exemplars demands human effort to refine retrieved triples. This can be expensive when scaling to many tasks or languages.
- **Computational overhead.** The retrieval, verification, and rethinking steps introduce additional computational cost compared with one-pass CoT prompting. Multiple inference rounds may be required for difficult queries.
- **Limited evaluation scope.** Experiments focus on English benchmarks and a specific family of LLMs. Generalization to other languages, domains, and architectures remains an open question.

These limitations suggest several future directions:

- Automating the construction of evidence triples using information extraction or joint training, thereby reducing human involvement.
- Extending Chain-of-Knowledge prompting to multi-hop, multi-modal, or interactive reasoning settings.
- Integrating reliability signals from F2-Verification into training objectives so that models learn to prefer factually correct and faithful reasoning paths.

7.3 Key Takeaways

The technical contributions of the paper can be summarized in three key points:

1. Representing reasoning evidence as explicit triples, together with textual explanations, produces a more verifiable and interpretable reasoning process than plain chain-of-thought text.

2. A combined view of factuality and faithfulness is essential for assessing the reliability of reasoning chains; considering only one of them is insufficient.

3. Iterative rethinking with knowledge injection enables LLMs to correct earlier mistakes and achieve stronger and more robust performance on diverse reasoning benchmarks.